

they immediately would release a new version of the product, which would go live on our website for a relatively small number of people. The team would be able immediately to assess the impact of their work, evaluate its effect on customers, and decide what to do next. For tiny changes, the whole process might be repeated several times per day. In fact, in the aggregate, IMVU makes about fifty changes to its product (on average) every single day.

Just as with the Toyota Production System, the key to being able to operate this quickly is to check for defects immediately, thus preventing bigger problems later. For example, we had an extensive set of automated tests that assured that after every change our product still worked as designed. Let's say an engineer accidentally removed an important feature, such as the checkout button on one of our e-commerce pages. Without this button, customers no longer could buy anything from IMVU. It's as if our business instantly became a hobby. Analogously to the Toyota andon cord, IMVU used an elaborate set of defense mechanisms that prevented engineers from accidentally breaking something important.

We called this our product's immune system because those automatic protections went beyond checking that the product behaved as expected. We also continuously monitored the health of our business itself so that mistakes were found and removed automatically.

Going back to our business-to-hobby example of the missing checkout button, let's make the problem a little more interesting. Imagine that instead of removing the button altogether, an engineer makes a mistake and changes the button's color so that it is now white on a white background. From the point of view of automated functional tests, the button is still there and everything is working normally; from the customer's point of view, the button is gone, and so nobody can buy anything. This class of problems is hard to detect solely with automation but is still catastrophic from a business point of view. At IMVU, our immune system is programmed to detect these business consequences and